

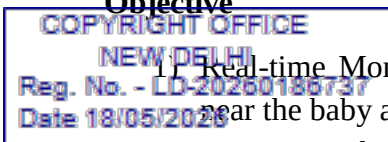
IDENTIFYING ABNORMAL INFANT CRY PATTERNS USING MACHINE LEARNING MODEL

Poonam Lambahte | Tanesh Devhare | Pranav Badole | Virendra Godse | Shrutam Hinge |

Introduction

Monitoring the different types of cries infant cries in babies from age group 0-2 years. The main reason for monitoring the cries is because many a times the parents are unable to differentiate between the normal cries and abnormal cries. Many times, by birth the infants can have some disorders related to neurological, behavioral disorders etc. which might in the near future may affect the infant. This toy will help in detecting the babies cries and the IoT device present inside the toy will perform the analysis of the baby cry and if the babies' cries are abnormal then an alert sound will be produced for the parents to check their baby's health. This is a precautionary device that will act as an alert system for the parent.

Objective



- 1) Real-time Monitoring: The IoT device is embedded inside the toy which will be present near the baby and if the baby cries, then it will monitor his cries instantly and give a real time outcome to the parents is it normal or abnormal cry.
- 2) Early Warning and Alerts: This device will act as a precautionary and an early warning system which will give an early alert to the parents about any kind of disease or any type of abnormalities and request the parent to consult a pediatric.
- 3) User-Friendly Interface: The user interface will use user-friendly with not much complication for the parents or user to understand more easily and without any confusion.
- 4) Power Efficient: The IoT toy has minimal requirements of electricity of it has a long-lasting battery and consumes less power and works on full efficiency.
- 5) Cost effective; The toy has a minimum requirement of gadgets and can be built easily so it is cost effective and can be bought by majority of the people as well as the needy ones.
- 6) Customizable according to the location: The dataset can be customized as per the different location of the toy where it will be used like in certain areas there are high occurring diseases then only those diseases will be given into the dataset of more efficient outcomes.

Problem formulation

In today's time there are many diseases that are emerging which are majority of the times neglected by the parents which will lead to some major problems further. Many a times these problems occur in the early stage like in the age of 0-2 and get neglected which will lead to major problem in the future. So, to provide a precautionary alert system for the parents about the baby this system is designed to tell if the cries of the baby are normal or not. And if the system detects any kind of abnormalities in the babies cries then it will instantly alert the parents and as to consult a pediatrician for cross verification and for further medications. This system will help to avoid and future diseases which later might be incurable.



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Methodology

In this project we will require hardware like ESP32 or Raspberry Pie which will act as a motherboard and provide the connection for all the other hardware devices. The motherboard will be fed with the data and the processed outcome. The baby cries will be analyzed when it cries, if the cry is normal then it will tell the parents that what shall be done to make them quiet and provide with a solution. If the cry is abnormal then it will alert the parents and ask to consult the doctors as soon as possible. There will be a transmitter which will be connected to the local network which will help to transmit the data from the toy to the parent mobile or any software where it is being integrated.

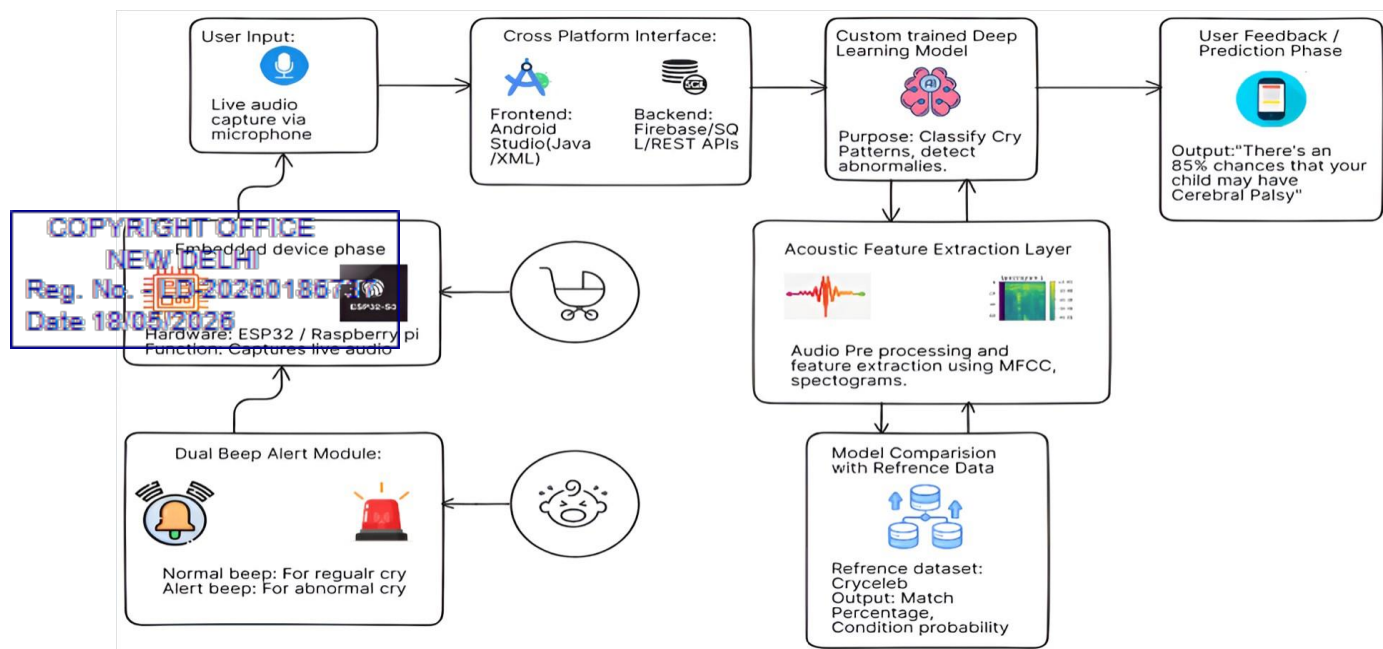


Fig 1. Architecture diagram of SmartCry system

1. Requirement analysis: We need to gather the voice notes of the various cries of the babies and then work on it using AI/ML and then the data will be ready for implementing into the toy
2. Development: The project is implemented using Python, AngularJS and NodeJS as all of these will result in efficient implementation without any problems and easy understanding.
3. System Design: The system is designed in such a way that it will be busily understood by every type of user.
4. Testing: The testing of the system can be done by using the recorded cries and cross verifying the system for its accuracy and correctness.
5. Deployment and Updates: The deployment of the system will be done into the toy and its updates can be done using the WIFI connection between the toy and the mobile network and the types of updates will be majorly on dataset and user interface.



Model

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1. Action: Extracting features like MFCC, FoGlide, Tgslide, RFsdys

Formula for Mel-Frequency Cepstral Coefficients (MFCC): Convert audio signals to frequency

domain using Fourier Transform.

$$X(f) = \sum_{n=0}^{N-1} x(n) \cdot e^{-j2\pi f n / N}$$

2. FOglide(Pitch Glide): Sudden change in fundamental frequency > 600 Hz indicates abnormality).

$$FO_{glide} = |f_{0_{n+1}} - f_{0_n}|$$

3. Tglide(Duration of pitch): How long the abnormal pitch lasted.

$$T_{glide} = t_{end} - t_{start}$$

4. Evaluation Metric: This is used to measure that the system false rejection and false acceptance errors are equal and lower the ERR value means more accurate and reliable is the system.

$$ERR = FAR = FRR$$

5. K-Nearest Neighbors (KNN): This algorithm compares the new baby cries with the known predefined baby cries.

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^n (x_{ik} - x_{jk})^2}$$

6. Decision tree: This is used to supervise the algorithm and splits the data into branches until it reaches to its final decision.

7. Random Forest: It combines multiple decision trees and comes up with a final output with improved accuracy and reduce overfitting.

$$\hat{y} = \text{mode}\{f_1(x), f_2(x), f_3(x), \dots, f_K(x)\}$$

8. Support Vector Machine: It separates the normal cry with abnormal cry most efficiently.

$$f(x) = w^T x + b$$

9. XG Boost: If there are any kind of errors in the decision trees the XG Boost will help to avoid it.

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i)$$



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Excepted Outcome:

The expected outcome of the above project is that it should tell rectify between normal cries and abnormal cries and tell the parents about it so proper steps will be taken to solve the babies' issues. Many a times these issues go unnoticed which will lead to major problems in the future and. This

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can be avoided and can save lives and its cost effectiveness will lead to help many families.

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